HybridQML · Clinical Diagnostic Telemetry

Hybrid Quantum-Classical ML Diagnostic Platform · SIH26139

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Attending Practitioner:	**Dr. Sarah Connor, MD**	**Inference Timestamp:**	`2026-08-28 14:37:42 UTC`
Diagnostic Dataset:	Breast Cancer Wisconsin (Diagnostic)	**Model Architecture:**	4-Qubit VQC (RY-RZ-CX) + Classical Baseline

DIAGNOSIS DETERMINATION: MALIGNANT (Disease-Positive / Critical)

Confidence Score: **91.4%** · Pathological Ground Truth: **Malignant** (MATCH ✓)

Dual Model Inference & Telemetry Breakdown

Model Branch	Evaluated Architecture	Predicted Output	Raw Score / Prob	Confidence
Quantum VQC	4 Qubits · 2 Layers (RY-RZ-CX Entanglement)	**Malignant**	`0.9140`	**`91.4%`**
Classical Baseline	Logistic Regression (Apples-to-Apples PCA)	**Malignant**	`0.9500`	**`95.0%`**

Biomarker Features & Explainability Attribution

Feature / Latent Dimension	Clinical Domain / Type	Ingested Value	Model Attribution (ΔAUC)
PC1	Quantum Latent Dimension	`0.3421`	+0.125 ΔAUC
PC2	Quantum Latent Dimension	`-0.1542`	+0.082 ΔAUC
PC3	Quantum Latent Dimension	`0.0891`	+0.045 ΔAUC
PC4	Quantum Latent Dimension	`-0.0125`	+0.012 ΔAUC